DOE SESSIONS 1-8 | PYTHON PROJECT

Smart Product Search

Data structures, recommendations, rules and analytics

ONE CONNECTED SYSTEM

Streamlit / CLI -> SearchEngine -> Catalog, graph and rules. Historical CSV -> Pandas / NumPy -> statistics and charts.

SEARCH AND ORGANIZE

Linked list and price BST; linear/binary name search; category and ID dictionaries; AVL rotations and educational B-tree splits.

RECOMMEND AND EXPLAIN

Manual min-heap, rating max-heap and deadline queue. BFS/DFS and weighted interactions. Greedy/A* traces, typed frames and eight prioritized discount rules.

DATA AND EVIDENCE

60 synthetic teaching products; 1,000 historical UCI products and 275,602 retained positive transaction lines. Categories are inferred; ratings are synthetic. Original prices and transactions are historical GBP values.

MEASURED TREE RESULT

For 20 prices inserted in sorted order, BST height was 20 nodes and AVL height was 5. This demonstrates balance; it is not a universal runtime speedup.

LEARNING AND LIMITS

Indexes improve particular queries but add update costs. Deletion rebuilds trees. Heuristic scoring preprocesses every product. UI edits are temporary; analytics uses the CSV snapshot. Peer review and live rehearsals remain student activities.

Source: Chen, D. (2015), Online Retail, UCI. doi.org/10.24432/C5BW33 | CC BY 4.0

Reproduce: python main.py --experiments | python main.py --analytics | See README.md